

# Thomas Russell

## AEROSPACE ENGINEERING AND SCIENCE GRADUATE

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### SKILLS & ABILITIES

**Software & CAD/CAM:** Siemens NX (+ Teamcenter), SolidWorks 2025, PTC Creo (+ Windchill PLM), Fusion 360 CAM, KiCad

**Simulation & DAQ:** ANSYS (FEA/CFD), NX Simcenter (NASTRAN), MATLAB & Simulink, Python, C++, C#, JavaScript, LabVIEW, LabJack

**Manufacturing:** Machining (CNC/Manual Milling & Lathe), Additive Manufacturing, Fabrication, Welding, Composites, Drawing (ASME Y14.5)

### PROFESSIONAL EXPERIENCE

#### Rocket Lab Development Engineer, Neutron Thrust Module

Summer 2025 – 14-week internship

- Executed structural analysis and sub-system testing on the thrust module to drive validation and flight qualification.
- Owned the complete component lifecycle, from conceptual design and CAD modeling to final drawing packages and manufacturing release. ([Read more](#))

#### Monash High Powered Rocketry (MHPR)

Since 2022

- Team Lead (2025–2026):** Directing 150+ students targeting a 100,000 feet launch and competing in the AURC 10k ft SRAD category; secured over \$85,000 in sponsorships.
- Propulsion Lead (2024–2025):** Led design, manufacture and test of 3kN ([Read more](#)) and 12kN (cryogenic liquid oxygen, [Read more](#)) hybrid engines, winning the IREC Jim Furfaro Award for Technical Excellence. Includes all ground support equipment and test infrastructure.
- Engineered composite over-wrapped pressure vessels, regeneratively cooled nozzles, and custom pneumatic oxidiser systems, taking hardware from concept to completion.

#### Hydrogen Jet Facility, Final Year Project

2026 – 2 semesters

- Design, build and commission of a hydrogen jet facility at Monash University. Responsible for implementing all fluids systems, control, data acquisition and safety measures for the facility.
- Investigating supersonic hydrogen jet leaks with acoustics involving the use of high speed schlieren imaging techniques and subsequent analysis. ([Read more](#))

#### Liquid Rocket Engine Project

2026 – Personal project

- Designed and built a nitrous oxide fill system including fluids, housing, custom PCB, complete networking and DAQ.
- Design and test of a N<sub>2</sub>O/IPA engine featuring a custom pyro-actuated Urbanski-Colburn valve and pintle injector.
- Characterised valve and injector empirical discharges. Developed a complete thermodynamic simulation for the entire engine to predict performance.
- Hot fire completed, exceeding performance expectations with C\* efficiency over 80%. ([Read more](#))

#### Summer Research Project, Shock Lab Research Group

Summer 2024 – 10-week program

- Designed, manufactured, and tested a planar glass rocket nozzle to investigate Free Shock Separation (FSS) phenomena.
- Executed complex CFD analysis and processed high-speed schlieren imaging data utilising Spectral Proper Orthogonal Decomposition (SPOD). ([Read more](#))

#### Assistant Robot Operator, Titomic

Since 2024

- Operating 6-axis robots equipped with cold spray additive manufacturing heads to produce commercial components.
- Conducting metallurgical R&D on novel alloys, preparing and analysing samples via high-powered microscopy to evaluate material properties.
- Machining product scaffolds, tensile coupons, and post-sprayed items using manual and CNC techniques.

### CERTIFICATES AND AWARDS

**Jim Furfaro Award for Technical Excellence, IREC 2025**  
**Runner up, 10k feet SRAD hybrid category, IREC 2025 –**  
**Machining short course, Chisholm Institute**  
**ANSYS FEA and CFD Introduction course, Leap Australia**

2025  
2025  
2023  
2023